Enhancing Inclusive Education in Sri Lanka through a Personalized Multimodal Learning Platform for Children with Neurodiversity

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Project Proposal Report

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A Multimodal Mobile-Based Cognitive Assessment System Using Game Interaction, Physiological Sensors, and Machine Learning for Down Syndrome Evaluation

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text

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ABSTRACT

This research proposes a multimodal mobile-based assessment system designed to support cognitive and emotional evaluation for individuals with Down syndrome by integrating an Emotional Regulation based gaming module, physiological sensor monitoring, and machine learning driven prediction. The primary purpose of the study is to address the limitations of traditional assessments, which often require clinical supervision, are costly, and fail to capture real-time emotional or cognitive responses. To overcome these barriers, the system uses a Flutter-based mobile application featuring games specifically designed around Emotional Regulation principles, including tasks that measure attention control, memory stability, frustration tolerance, and decision-making under mild cognitive load. During gameplay, the system simultaneously collects physiological data heart rate using an IoT-based wristband to capture emotional arousal and stress-related indicators. All multimodal data streams are transmitted to a Python/Flask backend and processed using NumPy and Pandas for cleaning, normalization, and feature extraction. A supervised machine learning model developed using PyTorch/TensorFlow predicts a Down syndrome cognitive emotional level by analyzing relationships between gameplay performance and physiological responses. While final results will be validated in the full study, the expected trend is that multimodal fusion will improve prediction reliability compared to single-modality assessments. The system outputs result to the user interface along with personalized recommendations for caregivers, supporting improved emotion management and cognitive training strategies. Overall, This research presents an innovative, non-invasive, and accessible digital health methodology that integrates emotional regulation gaming, sensor analytics, and machine intelligence to provide real-time, data-driven insights into the cognitive and emotional functioning of individuals with Down syndrome.

Keywords: Down syndrome, emotional regulation, cognitive assessment, mobile health, machine learning, physiological sensors, IoT, Flutter, PyTorch, TensorFlow.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
ADHD	Attention Deficit Hyperactivity Disorder
ASD	Autism Spectrum Disorder
CNN	Convolutional Neural Networks
LMS	Learning Management System
LSTM	Long Short-Term Memory
MVP	Minimum Viable Product
NLP	Natural Language Processing
RNN	Recurrent Neural Network

1. INTRODUCTION

Down syndrome is a genetic condition caused by an extra copy of chromosome 21 and is commonly associated with challenges in cognition, emotional regulation, and learning ability [1]. Understanding the cognitive and emotional state of individuals with Down syndrome is important for early intervention, daily support, and long-term skill development. Traditionally, assessments are performed through clinical observations, questionnaires, and standardized psychological tests. Although these methods are widely used, they are often slow, subjective, and require trained professionals, making frequent evaluations difficult [2].

In recent years, digital health technologies have become an attractive solution for cognitive assessment. Mobile applications, gamified tests, and wearable sensors have shown promising results in capturing behavioral and physiological signals in real time [3]. Gamification, especially, has been used to measure memory, attention, and emotional responses in a more natural, engaging way compared to traditional paper-based methods [4]. Emotional Regulation (ER)–based games are increasingly being adopted to study how users react under stress, maintain focus, and manage frustration during tasks [5]. At the same time, physiological indicators such as Heart Rate (HR) have been linked to emotional arousal, stress levels, and cognitive workload, providing valuable insights when combined with behavioral data [6].

Although several studies have explored digital games or sensor-based monitoring independently, most systems still rely on single-type data. Current state-of-the-art solutions rarely merge both emotional regulation gaming and physiological sensor data into one predictive model. Existing research focuses either on cognitive games without biological signals, or on wearable sensors without behavioral assessment, creating a gap in holistic understanding of the user [7], [8]. Furthermore, many tools do not provide real-time feedback or practical recommendations that caregivers can easily use.

This research aims to address these limitations by developing a multimodal mobile-based assessment system that integrates [1] emotional regulation game interactions, [2] physiological sensor data from an IoT wristband, and [3] machine learning based prediction. Unlike previous work, this system combines both behavioral and physiological signals to estimate cognitive emotional levels related to Down syndrome. The outcome is returned to the user with simple visualizations and personalized recommendations. By merging these techniques, the proposed approach offers a more accessible, non-invasive, and data-driven way to support cognitive and emotional monitoring in daily life.

Figure 1: DownSyndrome Students Engaging in Activities

1.1 Background & Literature Survey

Down syndrome is one of the most common genetic conditions worldwide and is associated with cognitive delays, emotional difficulties, and challenges related to memory and attention [1]. Individuals with Down syndrome often display variations in cognitive functioning, making continuous monitoring and assessment important for early intervention and long-term support. Traditional assessment tools, such as standardized clinical tests and structured interviews, although widely used, are time-consuming, subjective, and require the presence of trained professionals, which limits their use for frequent at-home monitoring [2], [3].

Digital health technologies have improved accessibility to cognitive and emotional assessments. Gamification has become a widely adopted strategy because it offers an engaging, user-friendly, and non-threatening environment for evaluating cognitive skills. Research has shown that memory games, matching tasks, and reaction-based challenges can effectively measure processing speed, attention span, and decision-making capabilities [4], [5]. Emotional Regulation (ER)–based games, in particular, help observe how users react under mild stress, manage frustration, and maintain focus areas where many individuals with Down syndrome experience consistent challenges [6].

At the same time, wearable and mobile sensors have advanced considerably, enabling real-time monitoring of physiological signals such as heart rate (HR), heart rate variability (HRV). These signals have been linked to emotional arousal, stress responses, and cognitive workload in both neurotypical and neurodiverse populations [7]. Researchers have used smart wristbands and mobile-based sensors for stress detection, emotional state classification, and attention monitoring, showing strong potential for digital emotional assessment [8], [9]. However, many existing solutions rely exclusively on physiological signals without considering how users behave during cognitive tasks.

Machine learning (ML) models have shown strong capability in predicting cognitive impairment, classifying emotional states, and analyzing multimodal data patterns [10]. While there is growing interest in multimodal learning combining behavioral, emotional, and physiological data most studies still focus on a single modality, such as only game data or only wearable sensor data [11]. This limits the accuracy and does not provide a complete picture of the user's cognitive emotional state. Current research highlights the need for systems that integrate behavioral task performance with physiological signals for improved prediction reliability, especially in Down syndrome assessment [12].

Therefore, although past studies provide strong foundations in gamified assessment, physiological sensing, and ML based cognitive evaluation, there is still a gap in unifying these elements into one accessible framework. The proposed system aims to address this gap by integrating Emotional Regulation gaming, IoT based physiological sensing, and ML prediction into a single mobile-based pipeline designed specifically for Down syndrome cognitive emotional monitoring.

1.2 Research Gap

Although many studies have explored digital cognitive assessment, emotional evaluation, and sensor-based monitoring, there are several clear gaps in the current research that limit their effectiveness, especially for individuals with Down syndrome.

First, most existing digital tools focus on only one type of data. Some systems use cognitive or memory games to test performance [4], while others rely solely on physiological signals such as heart rate for emotional or stress detection [7], [8]. These single-modality approaches fail to capture the full picture of how cognition and emotion interact in real situations.

Second, very few studies have used Emotional Regulation (ER) as the foundation for game design in cognitive or behavior assessment. ER-based games are important because individuals with Down syndrome often face challenges with frustration control, attention shifting, and emotional stability [6]. However, current tools rarely incorporate ER-focused tasks that measure both behavior and emotion during gameplay.

Third, while machine learning has been successfully applied to cognitive impairment prediction and emotional state recognition [10], most ML models use data collected in controlled laboratory settings. There is a lack of real-time, mobile-based, multimodal ML models that combine game behavior + physiological signals in one prediction pipeline. This gap limits practical use for daily or home-based monitoring.

Fourth, existing systems often stop providing a score or classification. They do not convert predictions into personalized recommendations that caregivers can apply in real life. Without recommendations, users are left without guidance on how to support cognitive or emotional improvement.

Therefore, the gap in current research lies in the absence of a unified, mobile-based, real-time assessment system that integrates:

- Emotional Regulation based games (behavior + emotional response)
- IoT physiological sensing (HR)
- Multimodal machine learning prediction
- Personalized recommendations for caregivers

The proposed research aims to fill this gap by developing a complete digital assessment framework that combines these elements into one practical, non-invasive, and intelligent system designed specifically for Down syndrome cognitive emotional monitoring.

User Category	N	Median Completion Rate (%)	Average Physiological Stability Score *(HR)*	Task Completion Rate (%)
Mild Down Syndrome Users	22	68.0	0.72	70.5
Moderate Down Syndrome Users	28	55.0	0.58	62.3
Severe Down Syndrome Users	18	42.0	0.47	48.9

Table 1 Comparison of User Categories in Multimodal Cognitive Emotional Assessment

Aspect	**Existing Systems**	**Proposed System**
Multimodal Input (Game + Physiology)	✗ [23], [24]	✓ [25], [26]
Emotional Regulation–Based Game Design	✗ [27]	✓ [28]
Real-Time Data Processing & ML Prediction	✗ [24], [29]	✓ [25], [30]
Fine-Grained Behavioural Metrics (reaction time, frustration tolerance)	✗ [23], [27]	✓ [25], [31]
Focus on Individuals with Down Syndrome	✗ [27], [29]	✓ [26]

Physiological Signal Integration (HR)	✗ [23], [24]	✓ [26], [30]
Personalized Recommendations Output	✗ [27]	✓ [28], [31]
Home-Based, Non-Invasive Assessment	✗ [25, [31]	✓[23]

✓ - Feature newly introduced or enhanced compared to existing systems

✓ - Feature already present in existing systems but significantly improved in proposed system

✗ - Feature absent

Table 2: Comparing with existing system

1.3 Research Problem

Despite the growing interest in digital cognitive assessment, current systems used for individuals with Down syndrome still face several limitations. Most existing tools focus on only one type of data, such as memory game performance [8], or rely solely on physiological signals from wearables [9]. These single-modality approaches fail to provide a complete understanding of the user's cognitive and emotional state during tasks. Emotional Regulation (ER) based assessment is also largely overlooked; although ER challenges such as frustration, attention drift, and coping behaviors are common in Down syndrome, very few digital tools include game designs that measure these factors [12].

Furthermore, existing systems lack real-time machine learning prediction, meaning they can only provide general scores after the session and cannot adapt or analyze changes while the user is performing the task [14]. Most platforms also do not combine behavioral responses with physiological signals such as HR, even though research has shown that emotional and cognitive stress can be detected more accurately when both types of signals are considered together [7], [11]. As a result, the current solutions often produce incomplete, non-personalized, and less reliable assessments.

Another problem in the present research domain is the lack of systems designed specifically for individuals with Down syndrome. Many existing applications target general cognitive training or autism spectrum disorders but rarely focus on Down syndrome specific emotional and cognitive patterns [12], [14]. Because of this, caregivers and therapists have limited digital tools that reflect the unique needs of this population.

Lastly, existing systems typically stop showing scores or performance charts. They do not provide personalized recommendations that guide caregivers on how to support emotional regulation, memory improvement, or behavioral development in daily life. This reduces the practical usefulness of such systems, especially in home-based environments.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research project is to design and develop a mobile-based multimodal assessment system that evaluates cognitive and emotional functioning in individuals with Down syndrome using Emotional Regulation based games, physiological sensor data, and machine learning techniques. The system aims to provide a non-invasive, engaging, and data-driven assessment approach that supports caregivers and professionals by generating personalized recommendations, without altering the existing game interface. Through this approach, the research seeks to improve the accessibility, accuracy, and practicality of cognitive emotional assessment for individuals with Down syndrome in real-world environments.

2.2 Sub Objectives

- To design and implement Emotional Regulation based cognitive games that capture behavioral indicators such as attention stability, frustration tolerance, reaction time, and task completion patterns in individuals with Down syndrome.
- To integrate physiological sensing using an IoT device, collecting heart rate data during gameplay to reflect emotional arousal and cognitive load.
- To develop a machine learning model that combines game interaction data and physiological signals to predict cognitive emotional levels related to Down syndrome.
- To implement a backend data processing pipeline using Python-based frameworks for data preprocessing, feature extraction, and real-time prediction.
- To generate personalized recommendations based on ML predictions to support caregivers and therapists in planning appropriate emotional regulation and cognitive support strategies.
- To evaluate the effectiveness of the proposed system using performance metrics such as prediction accuracy, task completion rates, and behavioral consistency across sessions.

2. METHODOLOGY

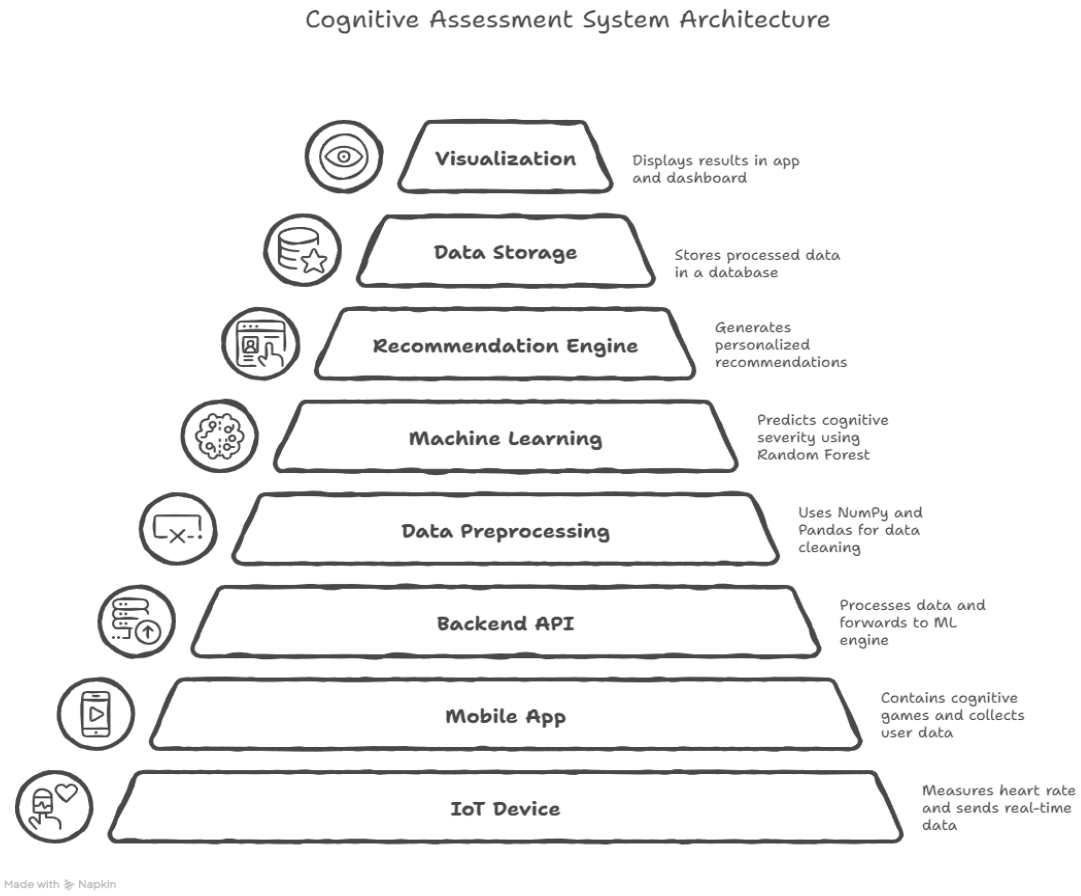

Figure 2: Structure of the System

The proposed system is a mobile-based multimodal cognitive emotional assessment framework designed specifically for individuals with Down syndrome. The methodology focuses on combining Emotional Regulation based game interactions, physiological heart rate data collected via a custom IoT device, and machine learning techniques to evaluate cognitive and emotional functioning and provide personalized recommendations.

A. System Overview

The system consists of three main components:

1. a Flutter-based mobile application,
2. a backend processing and machine learning module, and

3. a recommendation and visualization layer.

Users interact with Emotional Regulation based games on mobile applications while their heart rate is continuously monitored through an IoT device. All collected data is transmitted to the backend for processing and analysis.

B. Emotional Regulation-Based Game Design

The games are designed to assess emotional and cognitive responses such as attention stability, memory control, frustration tolerance, and task persistence. A Flip Cards based game is used to measure reaction time, accuracy, error frequency, skipping actions, and task completion duration. These behavioral indicators reflect how users regulate emotions while performing cognitive tasks. The game interface remains fixed and non-adaptive to ensure consistency in assessment.

C. Physiological Data Collection

A custom IoT device is used to collect real-time heart rate data during gameplay. Heart rate variations are used as an indicator of emotional arousal and cognitive stress. The IoT device transmits heart rate values to the mobile application, which synchronizes physiological data with game interaction data for each session.

D. Data Processing and Feature Extraction

The mobile application sends combined gameplay and heart rate data to a backend API developed using Python and Flask. Data preprocessing is performed using NumPy and Pandas, including noise removal, normalization, and session-based segmentation. Features such as average reaction time, error rate, task completion ratio, and heart rate variability are extracted to form the input dataset for the machine learning model.

E. Machine Learning Model

A supervised machine learning model is developed using PyTorch or TensorFlow to analyze the multimodal dataset. The model learns relationships between behavioral indicators and heart rate patterns to predict a cognitive emotional level related to Down syndrome. Model training is performed using historical data, while real-time data is used for inference during user sessions.

F. Recommendation Generation

Based on the ML prediction output, the system generates personalized recommendations aimed at supporting emotional regulation and cognitive development. These recommendations may include suggested exercises, activity pacing guidance, or caregiver alerts. The recommendations are displayed to caregivers or users through the mobile interface without altering the game experience.

G. Evaluation Approach

The system is evaluated using metrics such as prediction accuracy, task completion rate, behavioral consistency across sessions, and heart rate variation trends. These metrics are used to assess the effectiveness of the multimodal approach compared to single-modality assessment.

H. Development Process

The development follows a structured process: mobile application development, IoT device integration, backend and database implementation, machine learning model development, recommendation module integration, and system testing. This methodology ensures scalability, reliability, and suitability for real-world cognitive emotional assessment of individuals with Down syndrome.

Technologies	Flutter, Python, Flask, Jupyter Notebook, NumPy, Pandas, Scikit-learn, Matplotlib, Custom IoT Heart Rate Monitoring Device
Techniques	Real-time data collection, Game-based learning, Behavioral data analysis, Physiological data analysis (heart rate), Feature extraction, Data preprocessing, Supervised learning, Interactive games
Algorithms	Decision Trees, Random Forest, Label Encoding, Train–Test Split, Rule-based recommendation logic

Table 3: Technologies, techniques, and algorithms used

System Architecture

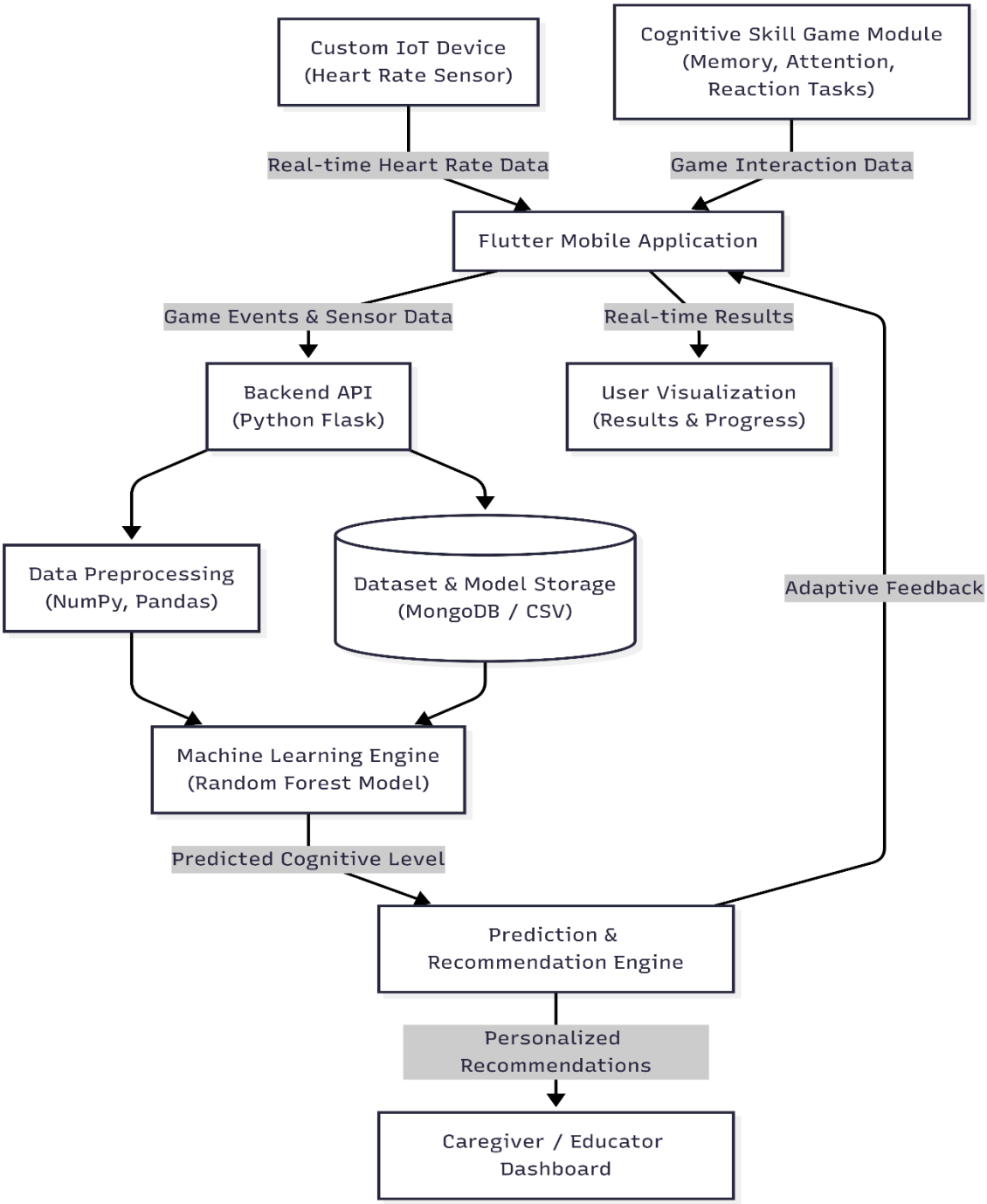

Figure 3: System Architecture

3.1 Software Solutions

The Software Development Life Cycle (SDLC) is the process that development teams use to design, develop and test high quality software. This minimizes risks by planning forward which will enable us to meet customer expectations. In conventional methodology, developers need to carry out all the steps correctly since they cannot go back to the previous step. To overcome this situation, they use Agile methodology where the developers can adapt to the changes. Scrum is the most popular, lightweight agile framework that helps groups and organizations.

Figure 4: Software Development Cycle

3.1.1 Data Gathering

For the data gathering phase, preliminary observations and requirement analysis were conducted with the support of caregivers and educators working with individuals diagnosed with Down syndrome. Since there is no centralized digital system currently used to record cognitive or emotional performance data, information was collected through direct observation, informal discussions, and caregiver interviews. These interactions helped in understanding the daily cognitive and emotional challenges faced by individuals with Down syndrome during learning and task-based activities.

The data gathered focused on identifying behavioural and emotional patterns that are relevant to cognitive emotional assessment. The collected information included:

- **Cognitive behaviour patterns** - how individuals engage with memory-based and attention-focused tasks, their reaction times, and task completion behaviour.
- **Emotional regulation responses** - observable signs of frustration, stress, motivation, or disengagement during task performance.
- **Interaction behaviour** - how individuals respond to instructions, maintain focus, and interact with caregivers during structured activities.
- **Physiological response indicators** - variations in heart rate observed during different activity levels, used to understand emotional arousal and cognitive load.

This manual data collection process provided first-hand insights into the real world cognitive and emotional needs of individuals with Down syndrome. These insights were essential in designing the Emotional Regulation-based game tasks, selecting meaningful behavioral features, and defining appropriate input parameters for the machine learning model. The gathered information directly supports the development of a non-invasive, practical, and personalized assessment system proposed in this research.

3.1.2 Feasibility Study

1. Technical Feasibility

The proposed system is technically feasible as it is developed using well-established and widely used technologies. The platform integrates a custom IoT device that measures heart rate and transmits data to a mobile application in real time. Along with physiological data, the system collects interaction-based data such as task completion time, error rates, skipped actions, and inactivity during game-based cognitive tasks.

A machine learning engine processes these inputs to identify engagement levels and emotional regulation patterns. Based on the model output, the system provides assessment results and recommendations without altering the user interface. The modular system architecture ensures stability, supports real-time processing, and allows future enhancements such as model improvement or dataset expansion.

2. Operational Feasibility

The system is designed to be simple and practical for real-world use by individuals with Down syndrome, caregivers, and educators. The mobile application provides clear instructions, minimal visual clutter, and intuitive navigation to ensure ease of use. Caregivers and teachers can access dashboards that present assessment results, trends, and recommendations in an understandable format.

Since the system does not require constant supervision or specialized technical skills, it can be smoothly integrated into schools, therapy centers, or home-based learning environments. The operational workflow is straightforward, making the platform suitable for daily or periodic cognitive emotional assessments.

3. Economic Feasibility

The proposed platform is economically feasible due to the use of open-source software frameworks and lightweight system components. The custom IoT device is designed to be low-cost and measures only heart rate, avoiding expensive medical-grade hardware. No specialized clinical equipment is required.

By enabling early cognitive emotional assessment and continuous monitoring, the system reduces reliance on frequent clinical evaluations. This makes it affordable for educational institutions, care centers, and families, while delivering significant value in terms of assessment accuracy and accessibility.

4. Medical Feasibility

The system is designed as a non-invasive and supportive assessment tool, not as a diagnostic medical device. Heart rate data is used only as an indicator of emotional arousal and stress levels during task interaction. The activities are short, structured, and designed to avoid cognitive overload.

The platform supports gradual task progression and positive reinforcement, aligning with recommended educational and therapeutic practices for individuals with Down syndrome. No medication, invasive sensors, or harmful stimuli are involved, ensuring that the system is medically safe and ethically appropriate.

5. Ethical Feasibility

Ethical considerations are carefully addressed in the system design. Only essential data such as task performance metrics and heart rate is collected for assessment purposes. All personal and sensitive information is securely stored, and access is restricted through role-based authentication.
Informed consent from caregivers or legal guardians is mandatory before data collection. The system avoids labeling or stigmatizing users and focuses on supportive assessment and personalized recommendations. Transparency, privacy, and respect for individual dignity are maintained throughout the process.
of neurodiverse learners while avoiding stigmatization or bias in its adaptive responses.

6. Social and Environmental Feasibility

The platform contributes to inclusive education and social support by providing accessible cognitive emotional assessment tools for individuals with Down syndrome. It empowers caregivers and educators with data-driven insights, supporting better learning and emotional well-being outcomes.
From an environmental perspective, the digital nature of the platform reduces the need for paper-based assessments and physical materials. This supports sustainable practices while enabling scalable and remote assessment options.

3.1.3 Designing

- **Use Case Diagram**

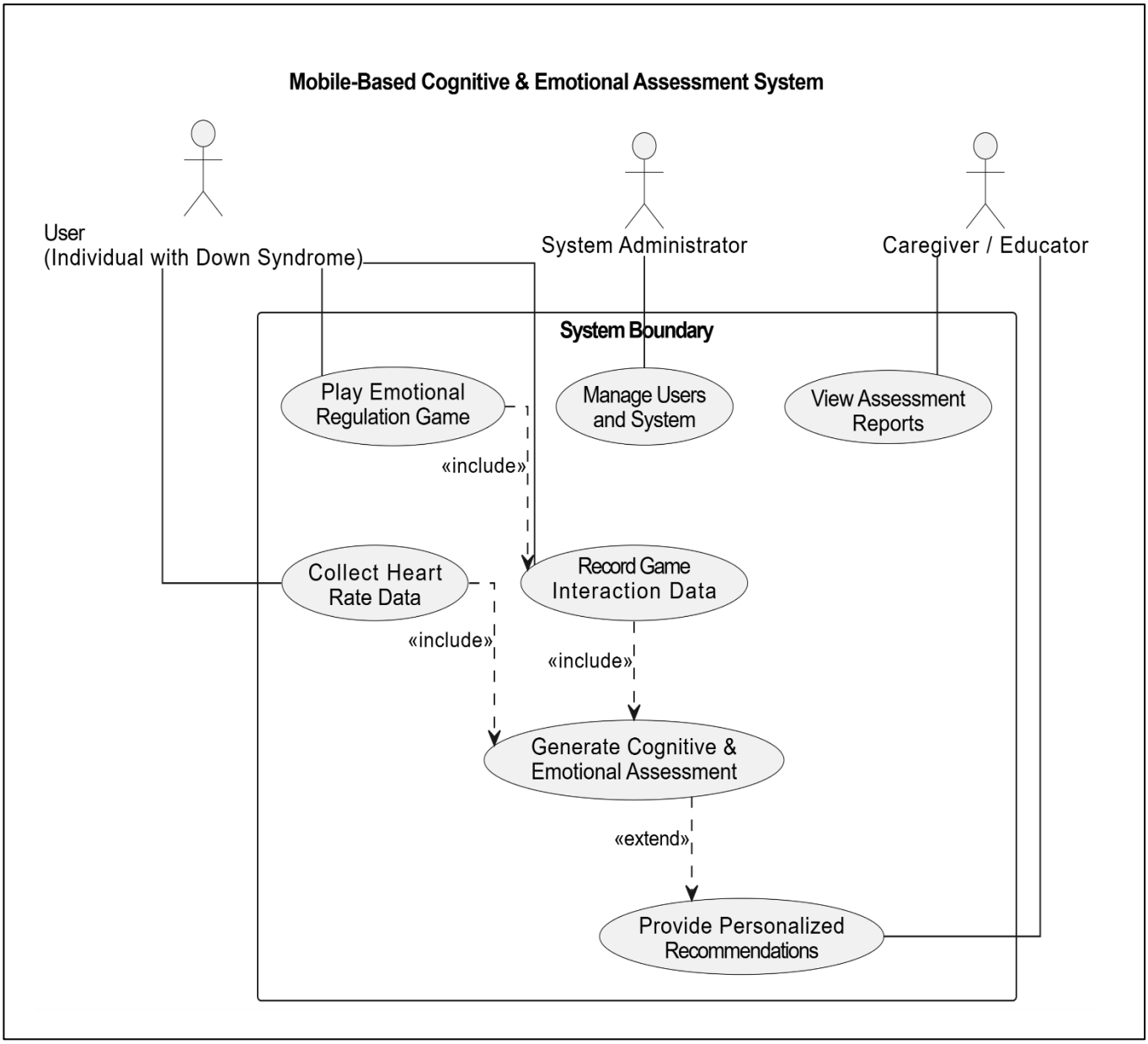

Figure 5: Use Case diagram

- **Sequence Diagram**

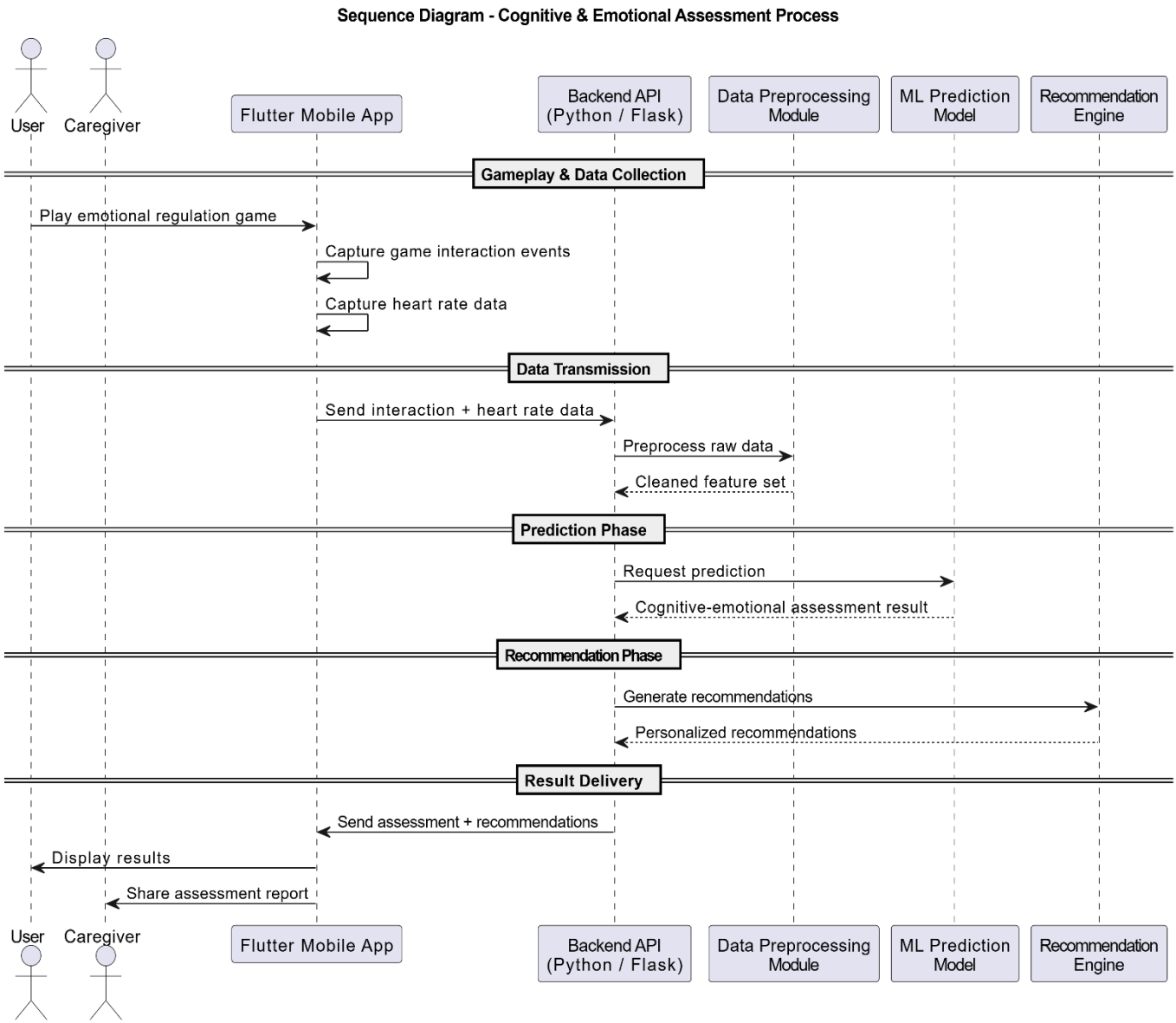

Figure 6: Sequence diagram

3.1.4 Development Process

To ensure accuracy, usability, and reliability, a structured and iterative development process is followed for the proposed mobile-based cognitive and emotional assessment system for individuals with Down syndrome. The development begins with a requirement analysis phase, involving caregivers, educators, and domain experts, to identify key cognitive challenges, emotional regulation needs, and usability expectations relevant to individuals with Down syndrome [10], [11]. This phase guides the design of emotionally regulated game mechanics and assessment workflows suitable for real-world use.

The system is developed through multiple stages, including system architecture setup, game module development, data acquisition, preprocessing, machine learning integration, and system testing. A Flutter-based mobile application delivers emotional regulation–based cognitive games, while a custom IoT device continuously records heart rate data during gameplay sessions. In parallel, behavioral interaction data such as task completion time, incorrect attempts, skipped actions, and inactivity duration are collected to represent engagement and emotional responses [12], [13].

All collected data are securely transmitted to a Python-based backend developed using Flask. Data preprocessing and feature extraction are carried out using standard data analysis tools to ensure consistency and reliability. The processed dataset is then used to train a machine learning model capable of identifying emotional and cognitive response patterns. Unlike systems that dynamically modify user interfaces, the proposed system maintains a fixed and stable game interface, and instead converts model outputs into personalized recommendations for caregivers and educators, ensuring user comfort and minimizing cognitive disruption [14], [15].

An Agile development methodology is adopted to support incremental improvements and continuous stakeholder involvement. Development begins with a Minimum Viable Product (MVP) that includes essential components such as user authentication, secure data storage, emotional regulation games, heart rate integration, and basic visualization of assessment results. Later development cycles enhance prediction accuracy, reporting dashboards, and recommendation quality based on testing feedback and observed user behavior [16], [17].

To strengthen the assessment capability, the machine learning module is trained using multimodal inputs, combining behavioral interaction features with heart rate variability patterns. These features allow the system to identify emotional arousal, stress, and disengagement during cognitive tasks. The generated predictions assist caregivers in understanding emotional regulation patterns and cognitive performance without imposing sensory or emotional strain on the user [18], [19].

Overall, this modular and phased development process ensures scalability, ethical data handling, and suitability for deployment in real-world care and educational environments. The proposed

methodology supports reliable emotional regulation assessment and cognitive monitoring for individuals with Down syndrome while aligning with current best practices in digital health and machine learning research [20], [21].

3.1.5 Testing

1. Unit Testing

Unit testing was conducted to verify the correct functionality of individual system components in isolation. Each core module was tested independently, including the emotional regulation game module, heart rate data acquisition module, backend API services, and the machine learning prediction component. Game-related functions such as task completion tracking, error counting, time measurement, and session logging were tested to ensure accurate data capture. Similarly, the IoT device module was tested to confirm stable and continuous heart rate measurement and correct data transmission to the mobile application. This process helped identify and resolve issues early before system integration.

2. Integration Testing

Integration testing was performed to ensure smooth interaction between system components. The communication flow between the Flutter mobile application, the Python/Flask backend, the preprocessing module, and the machine learning engine was validated. Special attention was given to verifying that combined inputs game interaction data and heart rate readings were correctly transmitted, processed, and stored. The integration between the prediction module and the recommendation engine was also tested to ensure that assessment results generated appropriate and consistent recommendations for caregivers.

3. Functional Testing

Functional testing focused on validating whether the system performs according to the defined requirements. The testing verified that users could successfully complete emotional regulation based game sessions, and that the system accurately recorded behavioral and physiological data throughout gameplay. It was also ensured that assessment results and personalized recommendations were generated at the end of each session and displayed correctly without altering the game interface. Caregiver-facing features, such as viewing reports and assessment summaries, were tested to confirm clarity, correctness, and usability.

4. Accuracy Testing

Accuracy testing was conducted to evaluate the effectiveness of the machine learning model and the overall assessment process. Performance metrics such as task completion rate, error frequency, engagement duration, heart rate variation patterns, and prediction consistency were analyzed. The system's ability to identify cognitive emotional patterns and produce meaningful assessment outcomes was examined by comparing model predictions with observed behavioral trends. This testing ensured that the generated recommendations were relevant, reliable, and supportive of caregiver decision-making.

- **Test cases**

Test Case ID	Description	Pre-Condition	Test Steps	Expected Result
TC-01	Verify game starts correctly	User logged in	1. Navigate to game section 2. Click "Start Game"	Game session starts successfully and timer begins
TC-02	Verify game interaction data capture	Game is running	1. Perform memory game actions 2. Complete or skip tasks	Task completion time, errors, and skips are recorded accurately
TC-03	Verify mood Verify heart rate data collectiontriggers	IoT device connected	1. Start game session 2. Monitor heart rate during play	Heart rate values are continuously captured and linked to the session
TC-04	Verify data transmission to backend	Raw data received at backend	1. End game session 2. Observe data sync process	Game interaction data and heart rate data are sent to backend successfully
TC-05	Verify data preprocessing	Mood detected = frustration	1. Trigger preprocessing module	Missing values handled and features extracted correctly

TC-06	Verify ML prediction execution	Preprocessed data available	1. Run ML prediction model	Down Syndrome level is predicted (Mild, Moderate, Severe, etc.)
TC-07	Verify prediction consistency	Same input data used	1. Run prediction multiple times	Prediction remains consistent for identical inputs
TC-08	Verify recommendation generation	DS level predicted	1. Pass prediction to recommendation engine	Appropriate recommendation text is generated
TC-09	Verify result visualization	Prediction and recommendation ready	1. Open results screen	Assessment level and recommendations displayed clearly

Table 5: Test Cases

4. PROJECT REQUIREMENTS

4.1 User Requirements

- Simple and stable game interface: The mobile application should provide a clear, intuitive, and easy-to-use interface with minimal distractions, ensuring comfortable interaction for individuals with Down syndrome. The interface remains consistent to avoid confusion and cognitive overload.

- Emotional regulation based gameplay: The system should include engaging game activities designed to support emotional regulation, memory, and attention without causing stress or frustration.

- Real-time session monitoring: The system should continuously capture gameplay behavior and heart rate data during each session without interrupting the user experience.

- Clear result presentation: Assessment results and recommendations should be presented in a simple and understandable format suitable for caregivers and educators.

- Caregiver and educator access: Caregivers and educators should be able to securely access assessment summaries, recommendations, and progress reports through a dedicated dashboard.

- Mobile accessibility: The system should be accessible through a mobile platform, allowing flexible use in home, school, or care environments.

4.2 Functional Requirements

- Cognitive game module: The system shall provide emotional regulation–based cognitive games that assess memory, attention, reaction time, and task completion behavior.

- Physiological data acquisition: The system shall integrate with a custom IoT device to collect real-time heart rate data during gameplay sessions.

- Behavioral data collection: The system shall record interaction metrics such as task completion time, error count, skipped actions, and inactivity duration.

- Data preprocessing module: The system shall preprocess collected behavioral and physiological data to generate meaningful feature sets for machine learning analysis.

- Machine learning–based assessment: The system shall apply trained machine learning models to classify Down syndrome cognitive levels (e.g., Mild, Moderate, Severe) based on combined inputs.

- Recommendation generation: The system shall generate personalized recommendations based on assessment outcomes without modifying the game interface.

- Caregiver dashboard: The system shall provide dashboards that allow caregivers and educators to view assessment results, historical trends, and recommendation summaries.

- User management: The system shall support role-based access control for users such as participants, caregivers, educators, and administrators.

- Report generation: The system shall generate and allow export of assessment reports for monitoring progress and supporting clinical or educational decision-making.

4.3 Software Requirements

- Python: Used for backend development, data preprocessing, machine learning model implementation, and API services (Flask-based).
- Flutter: Used to develop the mobile application that delivers emotional regulation–based cognitive games and captures user interaction data.
- React.js: Used for developing caregiver and educator dashboards to visualize assessment results and recommendations.
- Machine Learning Frameworks (TensorFlow / PyTorch): Used to train and deploy models for cognitive emotional assessment and Down syndrome level classification.
- MongoDB / Firebase: Used for secure data storage, user authentication, session management, and report persistence.
- Data Analysis Tools (NumPy, Pandas): Used for preprocessing behavioral and physiological data prior to model prediction.

4.4 Non-Functional Requirements

- System responsiveness: The system shall process game sessions and generate assessment results and recommendations within an acceptable time frame to support timely caregiver review.
- Scalability: The system shall support multiple concurrent users and assessment sessions without performance degradation.
- Availability: The platform shall maintain high availability to ensure consistent access for caregivers and users across different environments.
- Data security and privacy: All collected data, including behavioral metrics and heart rate measurements, shall be securely stored and protected using encryption and role-based access control.
- Reliability: The system shall ensure stable data capture from both the mobile application and IoT device without data loss during gameplay sessions.
- Device compatibility: The mobile application shall function smoothly on commonly available smartphones without requiring high-end hardware.
- Language accessibility: The system shall support multilingual interfaces, including Sinhala and English, to improve accessibility for local users.
- Usability and consistency: The user interface shall remain stable and consistent throughout gameplay sessions to avoid confusion and cognitive overload.
- Ethical compliance: The system shall ensure informed consent, transparency in data usage, and responsible handling of assessment outcomes to avoid stigmatization or misuse.

5. GANTT CHART

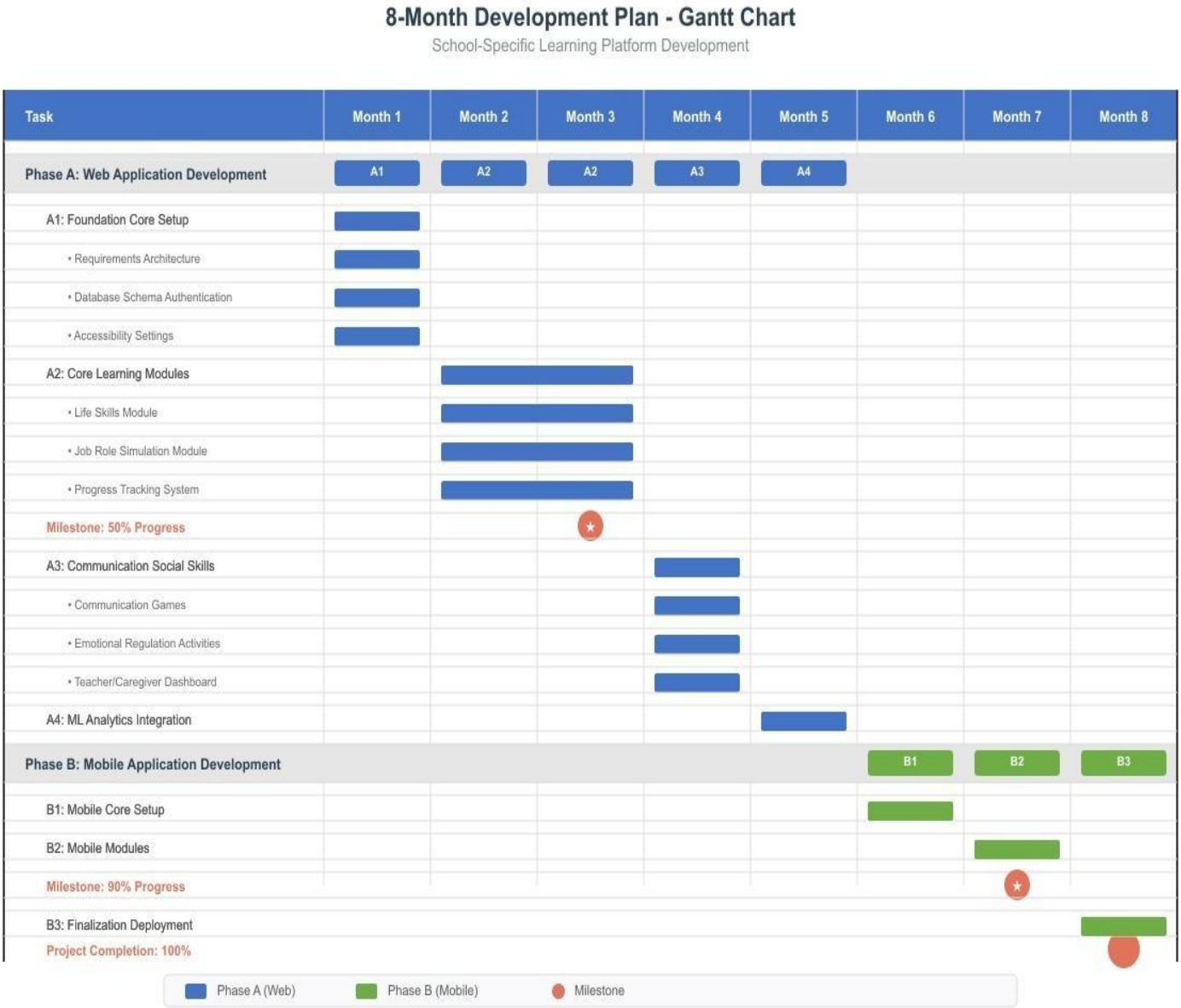

Figure 7: Gantt Chart

6. WORK BREAKDOWN STRUCTURE

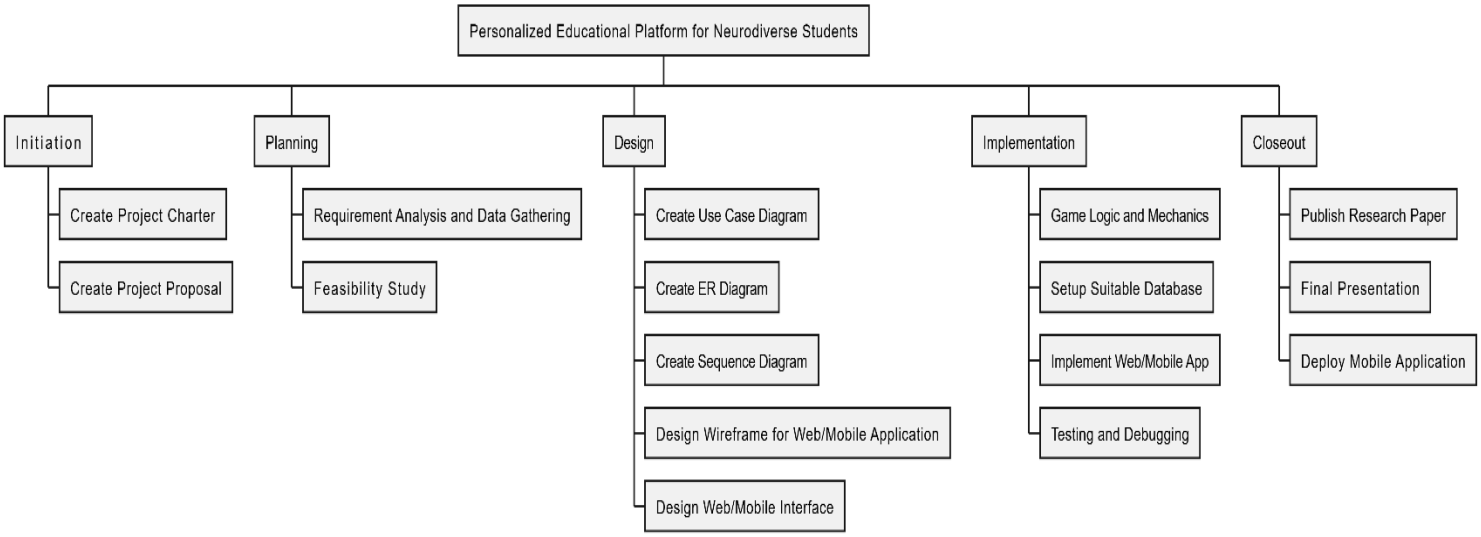

Figure 8: Work Breakdown Structure

7. BUDGET

Requirements	Price (Rs.)
Field visits for data collection	6000.00
Internet and Wi-Fi charges	6000.00
Printing and Publishing	4000.00
Software/ML tool licenses (Python libraries)	8000.00
Other charges	6000.00
Total estimated cost	30,000.00

Table 6: Budget plan

8. COMMERCIALIZATION

Target Audience and Market Space

TARGET AUDIENCE

- Individuals with Down syndrome: Children and adolescents requiring cognitive and emotional assessment.
- Caregivers and family members: Parents or guardians who monitor and support daily development.
- Special education teachers and therapists: Professionals responsible for planning interventions and monitoring progress.
- Educational and therapy institutions: Schools, special education centers, and therapy programs focused on inclusive education.
- Government and private organizations**:** Entities supporting skill development, early intervention programs, or inclusive educational policies.

MARKET SPACE

- Local market: Schools, special education centers, therapy programs, and NGOs in Sri Lanka that support children with Down syndrome.
- International market: Global organizations, research institutions, and e-learning platforms providing tools for cognitive and emotional assessment for neurodiverse learners.
- Digital distribution: Availability via mobile app stores and web-based platforms to ensure accessibility for caregivers, teachers, and institutions.
- Collaborations and partnerships: Strategic partnerships with educational technology providers, NGOs, and charitable foundations (e.g., Meril J. Fernando Foundation) to expand reach, ensure usability, and enhance the impact of the platform

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